Agentic AI-Based Intelligent Travel Planning System: A Multi-Agent Architecture for Context-Aware and Budget-Optimized Itinerary Generation

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***Abstract*—Travel planning often requires users to search across multiple independent platforms for transportation, accommodation, tourist attractions, restaurants, routes, weather conditions, and budget information. Existing travel platforms generally provide these services as separate functionalities, requiring users to manually collect and compare information and construct their own itineraries.**

This paper presents an Agentic AI-Based Intelligent Travel Planning System, a web-based application that uses a multi-agent architecture to coordinate different aspects of travel planning through a single intelligent interface. The system accepts natural language travel requirements, while a central Supervisor Agent decomposes the task into smaller subtasks and coordinates specialized agents such as the Transit Agent, Lodging Agent, Geospatial Agent, and Safety Agent.

The proposed architecture integrates open-source tools and public APIs including OpenStreetMap, OSRM, PostGIS, Nominatim, and Open-Meteo to support geographic reasoning, route computation, location-based search, and weather-aware planning. The Supervisor Agent validates budget and user constraints and generates a personalized day-wise itinerary.

Furthermore, the system incorporates dynamic replanning through LangGraph and local open-source large language models through Ollama to adjust travel decisions when conditions such as weather, availability, route changes, or budget constraints change. The proposed system therefore aims to provide an adaptable, budget-aware, geographically efficient, and personalized travel-planning experience.

***Index Terms*—Agentic AI, Multi-Agent Systems, Intelligent Travel Planning, Large Language Models, LangGraph, Ollama, Geospatial AI, Itinerary Generation, Constraint Optimization**

1. INTRODUCTION

Traditional travel planning tools and conversational chatbots typically operate through rigid scripts or single-shot response models, falling short when managing complex, multi-constrained, and real-world travel scenarios.

Real-world travel planning is not a single isolated decision, but a chain of interdependent choices spanning destination research, route selection, transportation, accommodation, points of interest, cost estimation, scheduling, and weather adaptation. A change in one decision may affect several other components of the itinerary. For example, selecting a distant accommodation can increase transportation costs and travel time, while unfavorable weather can require the replacement or rescheduling of outdoor activities.

The emergence of Agentic AI marks a fundamental paradigm shift from passive, reactive text generation to goal-directed, autonomous execution loops. Agentic systems combine language models with task-specific tools, persistent state management, planning mechanisms, and feedback-driven execution.

In travel planning, an agentic framework allows the system to autonomously break down vague user intentions into structured subtasks, execute tool calls, evaluate intermediate observations, and iteratively refine travel plans. This approach is particularly relevant because travel planning contains multiple interacting constraints involving time, cost, distance, user preferences, and environmental conditions.

Recent research demonstrates the importance of structured planning and tool-based reasoning in complex tasks. TravelPlanner shows that realistic travel planning requires simultaneous constraint satisfaction over long planning horizons [1]. ReAct further demonstrates that language models can improve task execution by alternating between reasoning and external

actions [2]. Similarly, Retrieval-Augmented Generation (RAG) provides a mechanism for grounding model responses using external information [3].

These developments suggest that a practical travel-planning system should not rely exclusively on a single language model. Instead, planning, retrieval, geographic processing, optimization, memory, and environmental adaptation should be coordinated as components of a larger intelligent architecture.

The proposed system follows this principle by introducing a central Supervisor Agent and several specialized agents. The Supervisor interprets the user's requirements and coordinates agents responsible for transportation, accommodation, geospatial processing, and safety. The architecture combines agentic reasoning with external geographic and environmental services to produce itineraries that are not only linguistically coherent but also constrained by practical travel conditions.

By utilizing local reasoning models such as Ollama and workflow orchestrators such as LangGraph, the proposed framework also aims to provide greater control over execution state, privacy, and dynamic replanning. The remainder of this paper reviews the existing research relevant to these design decisions and compares the major approaches that contribute to the proposed architecture.

2. LITERATURE REVIEW

The literature relevant to intelligent travel planning spans multiple research areas, including language-agent reasoning, retrieval-augmented generation, multi-agent coordination, optimization-based itinerary planning, recommender systems, persistent memory, web interaction, and adaptive autonomous systems. Rather than representing a single research direction, these studies collectively provide different components that can be integrated into an agentic travel-planning architecture.

2.1 TravelPlanner: Real-World Travel Planning and Constraint Satisfaction

J. Xie, K. Zhang, J. Chen, T. Zhu, R. Lou, Y. Tian, Y. Xiao, and Y. Su [1] introduced the TravelPlanner benchmark for evaluating real-world travel planning with language agents. The benchmark demonstrates that travel planning requires satisfying multiple simultaneous constraints across long planning horizons. The study further shows that standard large language models can struggle with constraint tracking, tool usage, and overall plan validity unless the planning workflow is properly structured.

The significance of this work for the proposed system lies in its emphasis on realistic constraints. A useful travel itinerary must simultaneously consider factors such as destination selection, transportation, accommodation, time, budget, and activity compatibility. Therefore, travel planning should be treated as a constraint-satisfaction and decision-making problem rather than merely a natural-language generation task.

2.2 ReAct: Reasoning and Acting in Language Models

S. Yao, J. Zhao, D. Yu, N. Du, I. Shafran, K. Narasimhan, and Y. Cao [2] proposed ReAct, a framework that combines reasoning traces with tool-based actions. The approach establishes an action-observation loop in which model-directed tool choices influence subsequent reasoning steps. This improves the ability of language models to interact with external environments while maintaining a reasoning process.

For travel planning, this reasoning-action paradigm is important because itinerary generation often requires external operations. The system may need to obtain geographic coordinates, calculate distances, search transportation options, retrieve weather information, and verify constraints before generating a final answer. Consequently, the proposed architecture adopts the general principle of combining reasoning with tool execution rather than generating an itinerary entirely from model knowledge.

2.3 Retrieval-Augmented Generation and Knowledge Grounding

P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Kuttler, M. Lewis, W.-t. Yih, T. Rocktaschel, S. Riedel, and D. Kiela [3] introduced Retrieval-Augmented Generation (RAG), which combines parametric language models with retrieved external information.

The central contribution of RAG is the ability to ground language generation using information obtained from external sources. This is particularly relevant to travel applications because travel information is dynamic and location dependent. A language model alone may not have current information about routes, weather, locations, or service availability.

In an agentic travel system, retrieval can therefore act as an information layer between the reasoning model and external services. The resulting architecture can use retrieved data as evidence for itinerary decisions instead of depending entirely on memorized information.

2.4 TravelAgent: Personalized Travel Planning

A. Chen, X. Ge, Z. Fu, Y. Xiao, and J. Chen [4] proposed TravelAgent, a travel-oriented assistant that integrates planning, recommendation, memory, and tool-usage modules.

The study highlights the importance of modular conversational architectures for improving personalization and comprehensiveness in intelligent travel planning. Travel preferences are inherently individualized; two users traveling to the same destination may prefer different activities, budgets, accommodation types, and transportation options.

This supports the use of specialized modules in the proposed architecture. Instead of treating all travel decisions as a single generation problem, the system separates responsibilities among agents and allows the Supervisor Agent to combine their results according to the user's requirements.

2.5 Graph-Based Multi-Agent Reinforcement Learning

J. Jiang, C. Dun, T. Huang, and Z. Lu [5] introduced graph convolutional structures for multi-agent reinforcement learning. The approach enables agents to exchange and aggregate information from neighboring nodes, supporting cooperation in multi-agent tasks.

The relevance of graph-based coordination extends beyond reinforcement learning. Travel planning naturally contains relationships between locations, routes, activities, transportation options, and temporal constraints. Consequently, representing travel information through connected entities can support more effective coordination between specialized planning components.

2.6 WebArena: Autonomous Interaction with Web Environments

S. Zhou, F. F. Xu, H. Zhu, X. Zhou, R. Lo, A. Sridhar, X. Cheng, T. Ou, Y. Bisk, D. Fried, and U. Alon [6] presented WebArena, a realistic web environment for evaluating autonomous agents interacting with APIs, databases, and dynamic web interfaces.

The study demonstrates that autonomous agents must do more than generate text. They must interpret observations, perform actions, handle intermediate states, and recover from failures. These capabilities are relevant to travel applications that may require interaction with several external services.

Although the proposed travel system primarily relies on public APIs and structured services rather than arbitrary web interfaces, the principles of tool interaction and environment-aware execution are directly applicable.

2.7 MemGPT: Long-Term Memory and State Management

C. Packer, S. Wooders, K. Lin, V. Fang, S. G. Patil, I. Stoica, and J. E. Gonzalez [7] proposed MemGPT, a memory management architecture that enables large language models to handle long-term context, operational state, and persistent storage.

Memory is important for travel planning because a planning process may involve several stages. User preferences, selected destinations, budget decisions, intermediate route calculations, and rejected alternatives may need to remain available throughout the planning process.

The proposed architecture therefore benefits from persistent workflow state. LangGraph can be used to maintain the execution state of the travel-planning process while specialized agents contribute information to the shared planning context.

2.8 AgentBench: Evaluation of LLM-Based Agents

X. Liu, H. Yu, H. Zhang, Y. Xu, X. Lei, H. Lai, Y. Gu, H. Ding, K. Men, K. Yang, and M. Zhang [8] introduced AgentBench, an evaluation framework for testing large language models as autonomous agents across interactive environments.

The framework emphasizes the importance of evaluating agents based on their ability to perform actions, interact with environments, handle errors, and complete tasks rather than evaluating only the quality of generated text.

For intelligent travel planning, this distinction is important. A fluent itinerary is not necessarily a valid itinerary. Evaluation must consider whether the proposed route satisfies time and budget constraints, whether locations are geographically feasible, and whether the sequence of activities is practically executable.

2.9 Multi-Agent Travel Planning

D. Camacho, D. Borrajo, and J. M. Molina [9] proposed a multi-agent travel planning system in which autonomous agents cooperatively search, filter, and reconstruct distributed travel information.

This work establishes travel planning as a multi-agent coordination problem rather than a single-agent information retrieval problem. Different travel services contain different types of information, and combining them requires coordination.

The proposed architecture follows a similar decomposition principle while incorporating modern language-model-based reasoning. The Transit Agent, Lodging Agent, Geospatial Agent, and Safety Agent can independently process their respective information before their outputs are coordinated by the Supervisor Agent.

2.10 Tourist Trip Design and Operations Research

J. Ruiz-Meza and J. R. Montoya-Torres [10] reviewed operations research models for personalized tourist trip design. The study emphasizes the Orienteering Problem, time-window constraints, spatial distances, and cost limitations as important components of itinerary optimization.

Operations-research approaches provide a mathematical foundation for itinerary generation. They are particularly useful when the objective is to select a set of activities while satisfying constraints on time, travel distance, and available resources.

However, conventional optimization methods generally require structured inputs. Natural-language user preferences are not always directly represented as mathematical constraints. This creates an opportunity for agentic systems in which language models interpret user requirements and optimization components validate or refine the resulting itinerary.

2.11 Mobile Recommender Systems in Tourism

M. Gavalas, C. Konstantopoulos, K. Mastakas, and G. Pantziou [11] surveyed algorithmic approaches and architectural frameworks for mobile tourist tour-planning systems.

The study highlights the importance of points-of-interest clustering, spatial routing, recommendation algorithms, and location-aware services. These capabilities are directly relevant to modern intelligent travel applications because the usefulness of an itinerary depends not only on which attractions are selected but also on how efficiently they can be visited.

The proposed system extends these concepts by combining spatial processing with language-based planning. Geographic services can provide objective distance and route information, while the agentic layer interprets preferences and coordinates the resulting information.

2.12 Sustainable and Autonomous Systems

S. Guenat et al. [12] discussed the role of robotics and autonomous systems in achieving sustainable development goals. The study highlights the potential of autonomous systems

for resource allocation, safety awareness, and environmental monitoring.

Although the study is broader than travel planning, it provides useful motivation for considering environmental and resource constraints in autonomous decision-making systems. Travel itineraries can involve transportation choices, resource consumption, safety conditions, and environmental considerations.

This suggests that future intelligent travel planners should not optimize only monetary cost and travel time but may also incorporate sustainability and environmental factors.

2.13 Agentic AI and Autonomous Intelligence

D. B. Acharya, K. Kuppan, and B. Divya [13] examined the foundational concepts of agentic artificial intelligence, goal-oriented architectures, and multi-agent methodologies.

The study provides a broader understanding of autonomous agents and their ability to pursue complex goals through interaction with their environment. This perspective is important for travel planning because the user's request can be expressed as a high-level goal rather than a predefined sequence of operations.

For example, a user may specify a destination, duration, budget, and preferred activities without explicitly describing how the itinerary should be constructed. An agentic system must therefore translate the high-level goal into intermediate objectives and actions.

2.14 Agentic AI and Large Language Model Architectures

S. Brohi, Q. A. Mastoi, N. Z. Jhanjhi, and T. R. Pillai [14] reviewed applications and structural patterns of agentic AI frameworks integrated with large language models.

The study highlights modular pipeline architectures and the increasing use of LLMs as reasoning components within larger systems. Such architectures are relevant to travel planning because different tasks require different capabilities.

A travel system can therefore combine an LLM-based Supervisor with specialized agents and external services instead of expecting one model to independently solve every aspect of the planning problem.

2.15 Decentralized and Distributed AI

L. Cao [15] examined decentralized artificial intelligence architectures and distributed processing models that can improve scalability and efficiency in large-scale applications.

The concepts of distributed intelligence and decentralized processing are relevant to multi-service travel applications. Transportation, accommodation, mapping, weather, and safety services may originate from different sources and may have different processing requirements.

The proposed system adopts a logically distributed architecture where specialized agents are responsible for different domains, while a central Supervisor coordinates their outputs. This provides modularity without requiring the complete system to be implemented as a fully decentralized network.

2.16 Socio-Cyber-Physical Systems and Dynamic Adaptation

M. Metta, S. Ciliberti, C. Obi, F. Bartolini, L. Klerkx, and G. Brunori [16] examined socio-cyber-physical systems that dynamically adapt to real-world operational changes and environmental feedback.

This perspective is relevant to travel planning because a generated itinerary may become invalid after environmental conditions change. Weather, route conditions, availability, or other external factors can affect previously selected activities.

Consequently, a practical travel-planning system should not treat itinerary generation as a single final operation. Instead, the system should maintain an executable planning state and support replanning when new observations invalidate existing decisions.

2.17 Synthesis of the Reviewed Literature

The reviewed studies reveal several complementary research directions. TravelPlanner and tourist trip-design research establish that travel planning is fundamentally a constraint-based problem involving multiple interacting variables [1], [10]. ReAct, RAG, and TravelAgent demonstrate how language models can be combined with external information, tools, and personalized planning mechanisms [2]–[4].

A second group of studies focuses on coordination and autonomous execution. Multi-agent travel planning and graph-based approaches demonstrate the benefits of distributing responsibilities across interacting components [5], [9]. WebArena and AgentBench further emphasize that autonomous agents must be capable of interacting with external environments, performing actions, handling intermediate observations, and recovering from execution problems [6], [8].

Another important research direction concerns persistent state. MemGPT demonstrates that long-term memory and operational state are important when language models participate in extended tasks [7]. In travel planning, this capability can preserve user preferences, intermediate decisions, constraints, and previous planning states across multiple execution steps.

The literature also demonstrates the importance of geographic and optimization components. Tourist trip-design research provides mathematical models for time windows, distances, costs, and activity selection, while mobile tourism recommender research emphasizes location-aware recommendation and spatial routing [10], [11]. These approaches complement language-model reasoning by providing objective geographic and optimization information.

Finally, broader research on agentic AI, distributed intelligence, sustainable autonomous systems, and socio-cyber-physical systems provides architectural foundations for autonomous systems capable of adapting to environmental changes [12]–[16].

2.18 Research Trends and Limitations

Despite the progress represented by these studies, the reviewed literature remains fragmented across different technical disciplines. Some studies primarily address language-model reasoning, while others focus on optimization, geographic

recommendation, memory, multi-agent coordination, or autonomous interaction.

This fragmentation creates a gap between intelligent conversational planning and executable travel planning. A language model may produce a plausible itinerary, but plausibility alone does not guarantee that the itinerary satisfies geographic, temporal, or financial constraints. Conversely, a conventional optimization algorithm can generate mathematically feasible routes but may struggle to interpret flexible natural-language preferences.

Another limitation is the handling of dynamic conditions. Travel information is not static, and changes in weather, route conditions, availability, or user preferences may require previously generated plans to be modified. Therefore, a practical system requires continuous state management and dynamic replanning rather than one-time itinerary generation.

The literature consequently motivates an integrated architecture that combines LLM-based reasoning with specialized agents, geospatial services, external data retrieval, constraint validation, persistent state, and dynamic replanning.

3. COMPARATIVE STUDY

The reviewed studies differ considerably in their objectives, inputs, computational techniques, and application domains. However, they collectively contribute important components toward the development of an intelligent travel-planning architecture.

A direct comparison is useful because no single study addresses all the requirements of a practical travel-planning system. Some approaches concentrate on reasoning and tool usage, whereas others focus on optimization, recommendation, multi-agent coordination, memory, or autonomous interaction. The proposed system aims to combine these complementary capabilities within a single architecture.

From the perspective of planning, TravelPlanner demonstrates the difficulty of satisfying multiple travel constraints simultaneously [1]. Operations-research approaches provide formal mechanisms for dealing with time windows, route distances, and activity selection [10]. These approaches are complementary: language agents can interpret flexible user requirements, while mathematical or geographic components can validate whether the resulting decisions are feasible.

From the perspective of agentic reasoning, ReAct establishes a reasoning-action cycle [2], while WebArena and AgentBench demonstrate the importance of interaction with external environments [6], [8]. These works indicate that a useful travel agent must be able to perform actions and evaluate their results rather than simply generate a textual response.

Multi-agent approaches provide another important architectural dimension. Graph-based multi-agent reinforcement learning and multi-agent travel planning demonstrate how specialized components can cooperate to solve complex tasks [5], [9]. This principle directly motivates the proposed division of travel planning into specialized Transit, Lodging, Geospatial, and Safety Agents.

Memory is another distinguishing factor. MemGPT demonstrates the importance of managing long-term context for extended tasks [7]. In the proposed system, persistent workflow state can allow the Supervisor Agent to retain constraints, intermediate results, selected alternatives, and replanning decisions.

Finally, broader agentic AI and distributed-system research provides architectural principles for scalable autonomous systems [13]–[15]. These principles support the use of a modular architecture in which individual services can be independently updated without redesigning the entire planning system.

TABLE I
COMPARISON OF SELECTED STUDIES – PART I

Ref.	Research Focus	Input	Main Technique	Main Limitation
[1]	Real-world travel planning with multiple simultaneous constraints	Travel requirements and planning constraints	Language-agent planning benchmark	Language agents may struggle with constraint tracking, tool usage, and overall plan validity
[2]	Reasoning and acting for decision-oriented tasks	User task and external tool observations	ReAct reasoning–action loop	Repeated reasoning and tool interaction can increase execution complexity
[3]	Knowledge-intensive generation and factual grounding	Language query and retrieved external information	Retrieval-Augmented Generation	Performance depends on the quality and relevance of retrieved information
[4]	Personalized intelligent travel planning	Travel requirements and user preferences	Planning, recommendation, memory, and tool-usage modules	Primarily focused on assistant-style interaction and personalization
[5]	Cooperation in multi-agent tasks	Graph-structured multi-agent information	Graph convolutional structures with reinforcement learning	Requires computationally intensive training and graph-based coordination
[6]	Autonomous interaction with realistic web applications	Web interfaces, APIs, and databases	Interactive web-agent evaluation environment	Complex web interactions can result in execution failures
[7]	Long-term context and operational state management	Long-term task context and stored information	Hierarchical memory management for LLMs	Requires careful memory management to maintain consistent long-term state
[8]	Evaluation of LLMs as autonomous agents	Interactive environments and agent actions	Multi-environment agent evaluation framework	Evaluation coverage may not directly represent specialized geospatial travel constraints

TABLE II
COMPARISON OF SELECTED STUDIES – PART II

Ref.	Research Focus	Input	Main Technique	Main Limitation
[9]	Distributed intelligent travel planning	Distributed travel information from web sources	Multi-agent search, filtering, and information reconstruction	Early architecture predates modern LLM capabilities and requires more manually defined coordination
[10]	Personalized tourist trip design and itinerary optimization	Tourist requirements, locations, time windows, and costs	Orienteering Problem and operations-research optimization	Exact optimization can become computationally expensive for large planning instances
[11]	Mobile recommender systems for tourism	Tourist preferences and points of interest	Mobile recommendation, POI clustering, and spatial routing	Traditional recommender architectures have limited conversational agent adaptability
[12]	Sustainable autonomous systems	Operational and environmental information	Robotics and autonomous-system approaches	Requires domain-specific customization for individual applications
[13]	Agentic AI and autonomous intelligence for complex goals	Goals, tasks, and environmental conditions	Agentic AI, goal-oriented architectures, and multi-agent methodologies	Broad focus without detailed specialization in travel planning
[14]	Agentic AI and LLM-based autonomous systems	LLM-based tasks and application requirements	Modular agentic AI pipeline architectures	High-level treatment with limited implementation detail for specific travel applications
[15]	Decentralized and distributed artificial intelligence	Distributed processing nodes and AI workloads	Decentralized AI and edge intelligence	Distributed components may introduce synchronization and coordination complexity
[16]	Socio-cyber-physical systems and real-world adaptation	Environmental and operational feedback	Integrated socio-cyber-physical system framework	Requires complex modeling of multiple interacting real-world variables

The comparisons presented in Tables I and II demonstrate that the reviewed approaches address different layers of the travel planning problem. The studies can broadly be divided into four groups: language-based reasoning and retrieval, multi-agent coordination, optimization and recommendation, and autonomous system infrastructure.

The first group, represented by TravelPlanner, ReAct, RAG, and TravelAgent, focuses on the ability of language models to understand requirements, reason over information, retrieve external knowledge, and interact with tools [1]–[4]. These approaches improve the flexibility of travel planning but do not by themselves guarantee geographically or mathematically valid itineraries.

The second group focuses on multi-agent coordination and autonomous interaction. Graph-based reinforcement learning and multi-agent travel planning demonstrate the value of cooperation between specialized components [5], [9]. WebArena and AgentBench extend this concept by emphasizing interaction with external environments and evaluation of agent behavior [6], [8].

The third group consists of optimization and recommendation approaches. Tourist trip-design research provides formal optimization mechanisms for selecting activities under temporal, spatial, and financial constraints [10]. Mobile tourism recommendation research further demonstrates the importance of points-of-interest selection and spatial routing [11]. However, these approaches generally assume structured inputs and therefore may require an additional interpretation layer when users express requirements in natural language.

The fourth group concerns system-level capabilities such as memory, sustainability, agentic architectures, distributed intelligence, and adaptation to environmental feedback [7], [12]–[16]. These studies contribute the infrastructure required for building systems that operate over longer periods and respond to changes in their environment.

3.1 Cross-Study Analysis

An important observation from the comparison is that information generation and constraint satisfaction are treated differently across the literature. Language-model-based systems are effective at interpreting natural-language requirements and generating flexible recommendations, whereas optimization-based approaches are stronger at enforcing explicit mathematical constraints.

This suggests that a practical travel-planning system should not attempt to replace optimization and geographic computation with language generation. Instead, the language model should act as a reasoning and coordination layer while specialized services perform objective calculations such as distance, route length, geocoding, and budget validation.

A second observation concerns dynamic information. Traditional recommendation and optimization approaches generally operate on a given set of inputs, whereas autonomous-agent research emphasizes interaction and adaptation. Real-world travel planning requires both capabilities because a plan can become invalid after new information is received.

A third observation concerns modularity. The literature provides evidence that different responsibilities can be separated among specialized components. This separation can improve maintainability because changes to routing, accommodation search, weather processing, or safety evaluation do not require redesigning the entire system.

3.2 Identified Research Gap

Although the reviewed studies provide strong individual contributions, there remains a gap in integrating their capabilities into a unified travel-planning workflow.

Existing travel-planning research demonstrates constraint-aware planning, but modern LLM-based reasoning and dynamic tool orchestration are not always integrated into the same architecture [1], [9], [10]. Conversely, general agentic AI research demonstrates reasoning, tool use, memory, and

autonomous execution, but is not specifically optimized for the geographic, temporal, financial, and safety constraints of travel planning [2], [6], [7], [13], [14].

Similarly, geographic recommendation and operations-research approaches provide valuable spatial and optimization capabilities, but they generally do not provide a natural-language interface capable of interpreting vague and evolving user goals [10], [11].

Therefore, the principal research gap is not the absence of individual planning, recommendation, optimization, or agentic techniques, but the lack of an integrated architecture that combines these capabilities into a single context-aware workflow.

The proposed system addresses this gap through a Supervisor Agent that coordinates specialized agents for transportation, accommodation, geospatial processing, and safety. The architecture also integrates external geographic and environmental services, persistent workflow state, constraint validation, and dynamic replanning.

4. DISCUSSION

The reviewed literature demonstrates that modern intelligent travel-planning systems require the integration of multiple technological components rather than relying on a single language model.

TravelPlanner establishes the importance of simultaneous constraint satisfaction in realistic travel scenarios, while ReAct demonstrates the effectiveness of combining reasoning and tool execution [1], [2]. Retrieval-Augmented Generation provides a mechanism for grounding language-model responses using external information sources [3]. TravelAgent further demonstrates the importance of modular planning, memory, recommendation, and tool-usage components [4].

Multi-agent approaches provide an effective architecture for decomposing travel-planning problems into specialized tasks. Graph-based approaches demonstrate how multiple agents can exchange information and cooperate [5]. The multi-agent travel-planning work of Camacho et al. further supports the use of distributed responsibilities for searching, filtering, and reconstructing travel information [9].

WebArena and AgentBench highlight the importance of evaluating agents in realistic interactive environments, while MemGPT provides mechanisms for persistent memory and long-term state management [6]–[8]. These capabilities are particularly important when a travel itinerary requires multiple planning iterations.

From the travel-planning perspective, operations-research approaches such as the Orienteering Problem provide mathematical foundations for route optimization, time-window constraints, distance minimization, and cost optimization [10]. Mobile tourism recommender systems further demonstrate the importance of spatial routing and points-of-interest recommendation [11].

The broader agentic AI literature suggests that autonomous systems should be modular, goal-oriented, adaptive, and capable of interacting with external environments [13], [14]. Distributed AI research additionally provides architectural principles for handling multiple computational components [15], while socio-cyber-physical system research emphasizes adaptation to real-world feedback [16].

Based on these findings, the proposed system adopts a hybrid architecture rather than relying exclusively on an LLM. The Supervisor Agent provides natural-language understanding, task decomposition, and coordination, while specialized agents perform domain-specific tasks. Geospatial tools provide objective location and routing information, external services provide environmental information, and constraint validation ensures that the generated itinerary remains within user-defined requirements.

This architecture creates a separation between reasoning and verification. The LLM can determine what should be considered, while deterministic tools can determine whether the resulting decision is geographically or computationally feasible. Such a separation is important for reducing unsupported recommendations and improving the reliability of generated itineraries.

Dynamic replanning further extends this architecture. Instead of treating an itinerary as a static document, the system maintains the planning state and can reconsider previous decisions when new information becomes available. This capability is particularly relevant to weather-sensitive activities, route changes, budget variations, and other real-world travel conditions.

5. CONCLUSION

The reviewed literature establishes a strong multidisciplinary foundation for the development of an Agentic AI-Based Intelligent Travel Planning System.

Research in multi-agent task decomposition and reasoning-action frameworks confirms that distributing travel-planning responsibilities across specialized agents can improve the handling of complex tasks compared with conventional monolithic chatbots [2], [5], [9]. TravelPlanner and tourist trip-design research further establish the importance of simultaneous constraint satisfaction involving time, distance, cost, and user requirements [1], [10].

Studies on retrieval, memory, autonomous interaction, and agent evaluation demonstrate that an intelligent system must also be capable of obtaining external information, maintaining state, performing actions, and evaluating intermediate results [3], [6]–[8].

Furthermore, research in tourism recommendation and geographic routing validates the importance of spatial information in itinerary generation [11]. Broader work on agentic AI, distributed intelligence, sustainability, and adaptive socio-cyber-physical systems provides architectural concepts for developing systems capable of operating under changing environmental conditions [12]–[16].

Based on these findings, the proposed system combines a central Supervisor Agent with specialized Transit, Lodging, Geospatial, and Safety Agents. The architecture integrates external services such as OpenStreetMap, OSRM, PostGIS,

Nominatim, and Open-Meteo with LangGraph-based workflow orchestration and local language models through Ollama.

The resulting architecture is intended to provide a more context-aware, budget-conscious, geographically efficient, and adaptable travel-planning workflow. Rather than treating itinerary generation as a single language-generation task, the proposed approach treats travel planning as a coordinated process involving reasoning, retrieval, geographic computation, constraint validation, memory, and dynamic replanning.

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